Slide Set 1

Noise

Danger	0	?	1	Danger

Vmin V0 VIL VIH V1 Vmax

-0.3V 0.7V 2.5V

0V 1.7V 2.8V

Verilog:
- HDL
- Describes logic gates
- used as input to synthesis/simulation

Combinational Logic
- only depends on present inputs
- No memory

Sequential Logic
- Has memory
- includes feedback
- Synchronous uses a clock

Boolean Properties

Communicative	$X \wedge Y = Y \wedge X$	$X \vee Y = Y \vee X$
Associative	$X \wedge (Y \wedge Z) = (X \wedge Y) \wedge Z$	$X \vee (Y \vee Z) = (X \vee Y) \vee Z$
Distributive	$X \wedge (Y \vee Z) = (X \wedge Y) \vee (X \wedge Z)$	$X \vee (Y \wedge Z) = (X \vee Y) \wedge (X \vee Z)$
Idempotence	$X \wedge X = X$	$X \vee X = X$
Absorption	$X \wedge (X \vee Y) = X$	$X \vee (X \wedge Y) = X$
Combining	$(X \wedge Y) \vee (X \wedge \bar{Y}) = X$	$(X \vee Y) \wedge (X \vee \bar{Y}) = X$
DeMorgan's	$\overline{(X \wedge Y)} = \bar{X} \vee \bar{Y}$	$\overline{(X \vee Y)} = \bar{X} \wedge \bar{Y}$

Axioms

$0 \wedge X = 0$ $1 \wedge X = X$ $\bar{0} = 1$

$1 \vee X = 1$ $0 \vee X = X$ $\bar{T} = 0$

Identity Idempotence negation

Sum of Products Normal Form

Product term, corresponds to a row in truth table with output = 1

$f(a,b,c) = (\bar{a} \wedge b \wedge c) \vee (a \wedge \bar{b} \wedge c) \vee (a \wedge b \wedge \bar{c}) \ldots$

Dual Functions

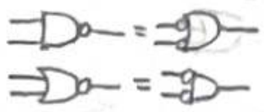
$f^D(\bar{a}, \bar{b}, \ldots) = \overline{f(a,b,\ldots)}$

Switch:

$\wedge - \vee$	$0 - 1$
$\vee - \wedge$	$1 - 0$

*Do not change complements

DeMorgans Law



Slide Set 2

Verilog literal numbers

Boolean → Verilog: <size>'[<signed>]<radix>value

ex: 8'sb0001_1101

- S, it ommited unsigned

& AND

| OR

~ NOT

^ XOR

Replicator {k{n}}

Wire[1:0]X = 2'b10;

Wire[7:0]y = {4{x}};

Cool Things

2Darray: //y = 8'b10101010

reg[3:0] data[7:0];

//Data contains 8 elements each with 4 bits

Bitwise expression:

Wire[3:0]result = A & B;

//result[0] = A[0] & B[0].....

Bitwise of one element:

wire result = |A;

//result = (A[0]|A[1]|A[2].....)

Binary	one-hot
000	00001
001	00010
010	00100
⋮	⋮

Named Port Association *order doesn't matter, connect directly

Mux3 #(16) m1(a2, a1, a0, s, b);

Mux3 #(16) m1(.a(a0), .b(a1), .c(a2), .sel(s), .out(b));

Test Bench 1

```
module test_maj;
  reg[2:0] count;
  wire out;
  Majority m(count[0], count[1], count[2], out);
  initial begin
    count = 3'b000;
    repeat(8) begin
    #100;
    $display("in = %b, out = %b", count, out);
    count = count + 3'b001;
    end
  end
endmodule
```

Test Bench 2

```
module q7_tb;
  reg[2:0] A;
  reg B,C,D,E;
  wire F;
  q7 DUT(A,B,C,D,E,F);
  initial forever begin
    E=0; #5;
    E=1; #5;
  end
  initial begin
    A=3'b100; B=0; C=0; D=1; #10;
    A=3'b111; D=0; #10;
    C=1; B=1; #10;
    D=1; #10;
    $stop;
  end
endmodule
```

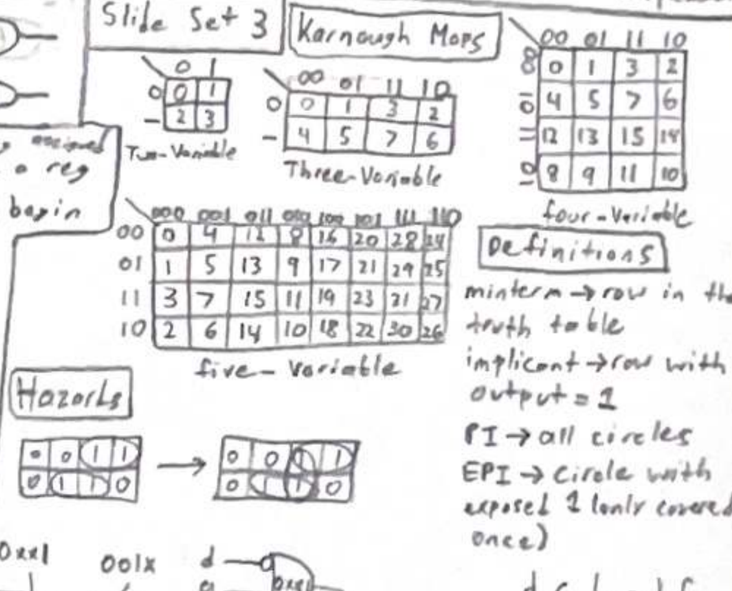
Always Block *anything assigned must be a reg

```
always @(sel or a or b) begin
  //order matters
end
```

Rules:
1. Every input is in sensitivity list or @(*)
2. Must have an output for every combo of inputs or default:

Slide Set 3

Karnaugh Maps



Definitions

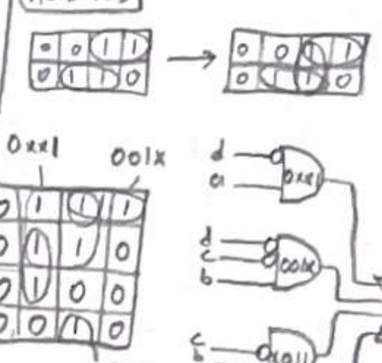
minterm → row in the truth table

implicant → row with output = 1

PI → all circles

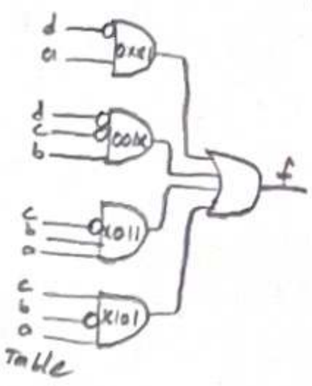
EPI → circle with exposed 1 (only covered once)

Hazards



$f = (a \wedge \bar{d}) \vee (b \wedge \bar{c} \wedge \bar{d}) \vee (a \wedge \bar{b} \wedge c) \vee (a \wedge b \wedge \bar{c})$

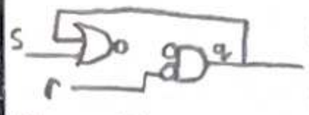
Product of Sums → Focus on 0's in Truth Table



d	c	b	a	f
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

Slide Set 4/5

Reset-Set (RS) Latch



r=1 → q=0

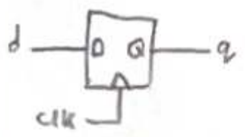
r=0, s=1 → q=1

r=0, s=0 → q holds last value

$q_{new} = \bar{r} \wedge (s \vee q_{old})$

↓ equivilant

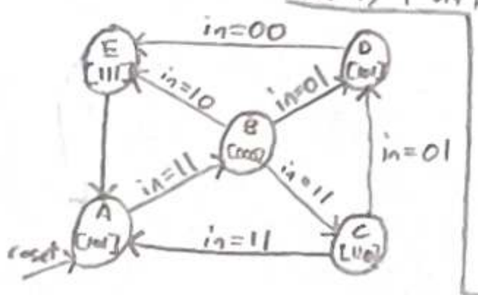
D Flip-Flop



- q = d on clk rising edge

n-bit register?
- Flip flop for each bit all connected to same clk

```
module FSM(clk, reset, in, out);
  input clk, reset;
  input[1:0] in;
  output[2:0] out;
  `define Sa 3'b000
  `define Sb 3'b001
  `define Sc 3'b010
  `define Sd 3'b011
  `define Se 3'b111
  Wire[2:0] present_state, state_next_reset, state_next;
  reg [6:0] next;
  VDFF #(3) STATE(clk, state_next_reset, present_state);
  assign state_next_reset = reset ? `Sa : state_next
  always @(*) begin
    casex({in, present_state})
      {2'b11, `Sa}: next = {`Sb, 3'b101};
      {2'bxx, `Sa}: next = {`Sa, 3'b101};
      {2'b10, `Sb}: next = {`Sc, 3'b000};
      {2'b01, `Sb}: next = {`Sd, 3'b000};
      {2'b11, `Sb}: next = {`Sc, 3'b000};
      {2'bxx, `Sb}: next = {`Sb, 3'b000};
      ⋮
      default:  next = {`Sa, 3'b101};
    endcase
  end
  assign {state_next, out} = next;
endmodule
```

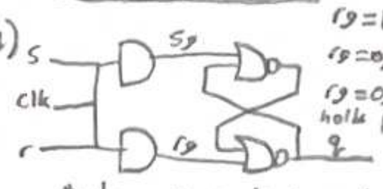


Gated RS Latch

rg=1 → q=0

rg=0, sg=1 → q=1

rg=0, sg=0 → q holds last value



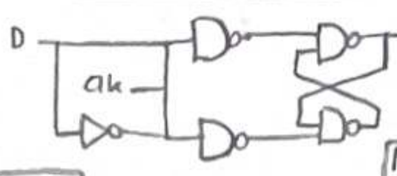
*when clk is high, acts as a RS latch

*when clk is low, maintains value regardless of s, r

ie: level sensitive

```
module vDFF(clk, in, out);
  parameter n=1;
  input clk;
  input [n-1:0] in;
  output[n-1:0] out;
  reg [n-1:0] out;
  always @(posedge clk)
    out = in;
endmodule
```

D Latch



clk=high → q = D

clk=low → q maintains

Non Blocking Assignment
- Signals are updated at the same time at the end of the always block (<=)

Shifter x<<y
- Shift x to the left by Y bit positions

Decoder

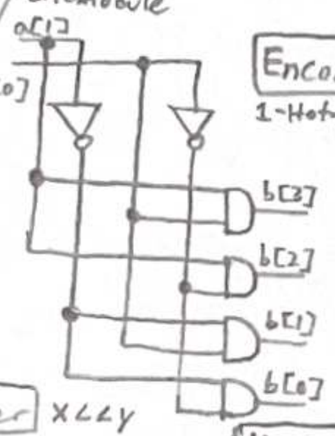
```
module Dec(a,b);
  parameter n=2;
  parameter m=4;
  input[n-1:0] a;
  output[m-1:0] b;
  wire[m-1:0] b = 1<<a;
endmodule
```

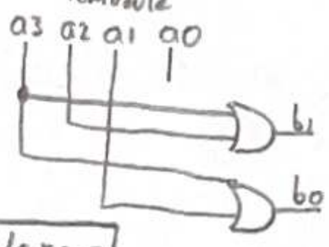
Binary → 1-Hot

Encoder

```
module Enc42(a,b);
  input [3:0] a;
  output [1:0] b;
  assign b = {a[3]|a[2], a[3]|a[1]};
endmodule
```

1-Hot → Binary

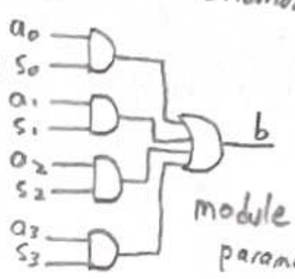




Multiplexer

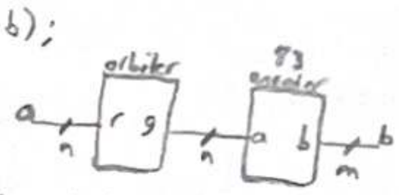
4input k wide mux

```
module Mux4(a3,a2,a1,a0,s,b);
  parameter k=1;
  input[k-1:0] a0,a1,a2,a3;
  input[3:0] s;
  output[k-1:0] b;
  reg[k-1:0] b;
  always@(*) begin
    case(s)
      4'b0001: b = a0
      ⋮
      4'b1000: b = a3
      default: b = {k{1'bx}};
    endcase end endmodule
```



Priority Encoder

```
module PE(r,b);
  input[7:0] r;
  output[2:0] b;
  wire[7:0] g;
  Arb #(8) a(r,g);
  Enc83 e(g,b);
endmodule
```



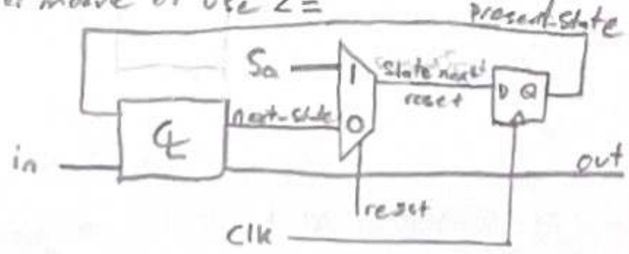
- Finds position of first 1 in a and outputs binary of that position

Purely Synchronous
1. only clk in sensitivity list
2. only signals that change on the same edge in the same always block
3. clk should not be read/written inside
4. Block can only describe what happens in 1 clk cycle
5. only one purely sync per module or use <=

Purely Level Sensitive (will give inferred latch)

```
always@(*)
  if (clk==1'b1)
    z = A&B;
```

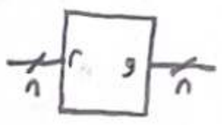
1. use only (*) for sensitivity list
2. clk only in 1 if condition
3. Same as ← 2 (PS)
4. Same as ← 3 (PS)

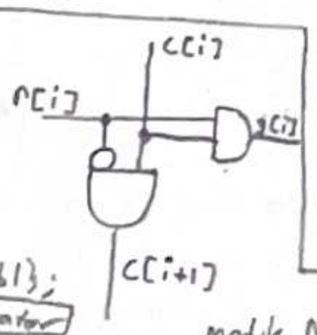


Slide Set 6/7

Arbiter
- Finds a 1 starting at LSB and outputs a one hot-code as the position of that one

```
module Arb(r,g);
  parameter n=8;
  input[n-1:0] r;
  output[n-1:0] g;
  wire[n-1:0] c;
  wire[n-1:0] g;
  assign c = {(~r[n-2:0] & c[n-2:0]), 1'b1};
  assign g = r & c;
endmodule
```



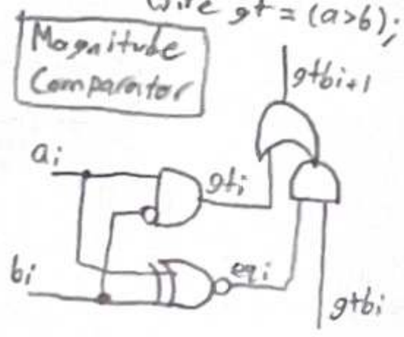


Equality Comparator

```
module Eqcomp(a,b,eq);
  parameter k=8;
  input[k-1:0] a,b;
  output eq;
  wire eq;
  assign eq = (a==b);
endmodule
```

```
module MagComp(a,b,gt);
  parameter k=8;
  input[k-1:0] a,b;
  output gt;
  wire gt = (a>b);
```

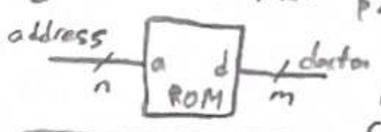
Magnitude Comparator



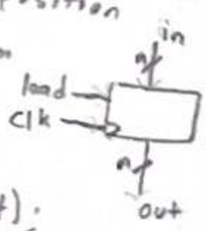

```
module VDFFE(clk, en, in, out);
  parameter n=1;
  input clk, en;
  input[n-1:0] in;
  output[n-1:0] out;
  reg[n-1:0] out;
  wire[n-1:0] next_out;
  assign next_out = en? in:out;
  always @(posedge clk)
    out = next_out;
endmodule
```

Register with load enable

Read Only Memory




```
module ROM(read_address, dout);
  parameter data_width = 32;
  parameter addr_width = 4;
  parameter filename = "data.txt";
  input[addr_width-1:0] read_address;
  output[data_width-1:0] dout;
  reg[data_width-1:0] mem[2**addr_width-1:0];
  initial $readmemb(filename, mem);
  assign dout = mem[read_address];
endmodule
```



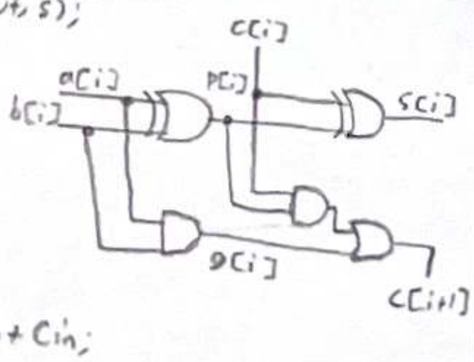
2's Complement

Steps:
1. Flip all bits
2. Add one

-8	4	2	1	
1	0	0	0	-8
1	0	0	1	-7
1	0	1	0	-6
⋮				
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3

Multi-Bit adder

```
module Adder1(a,b,cin,cout,s);
  parameter n=8;
  input[n-1:0] a,b;
  input cin;
  output[n-1:0] s;
  output cout;
  wire[n-1:0] s;
  wire cout;
  assign {cout,s} = a+b+cin;
endmodule
```



Pipeline Cycles

```
LDR r1, [r2, #4]      F D X M W
STR r1, [r1, #8]        F D S X M W
ADD r3, r3, r1            F D S X M W
```

```
module RAM(clk, read_address, write_address, write, din, dout);
  parameter data_width = 32;
  parameter addr_width = 4;
  parameter filename = "data.txt"
  input clk;
  input [addr_width-1:0] read_address, write_address;
  input write;
  input [data_width-1:0] din;
  output [data_width-1:0] dout;
  reg [data_width-1:0] dout;
  reg [data_width-1:0] mem [2**addr_width-1:0];
  initial $readmemb(filename, mem);
  always @(posedge clk) begin
    if(write)
    mem[write_address] <= din;
    dout <= mem[read_address];
  end
endmodule
```

```
module AddSub(a, b, sub, s, ovf);
  parameter n=8;
  input [n-1:0] a, b;
  input sub; //subtract if sub=1 else add
  output [n-1:0] s;
  output ovf;
  wire c1, c2;
  wire ovf = c1 ^ c2;
  Adder1 #(n-1) a1(a[n-2:0], b[n-2:0] ^ {n-1{sub}}, sub, c1, s[n-2:0]);
  Adder1 #(1) a2(a[n-1], b[n-1] ^ sub, c1, c2, s[n-1]);
endmodule
```

Note:
STMFD r13!, {r0-r7, r14}
push {R0-R7, lr}

Slide Set 8

N-Type: gate, drain, source
P-Type: gate, source, drain
① Gate = 0V → open ① Gate = 0V → closed
② Gate = 2.5V → closed ② Gate = 2.5V → open

NOT Gate: in, out, 2.5V

PFET → Pull-up
NFET → Pull-down

NAND Gate: a, b, f

NOR Gate: a, b, f

2.5V, 2.6V, 2.5V

Tri-State Inverter: a, e, f
- used if want to drive single wire from two or more places

e	a	f
0	0	Z
0	1	Z
1	0	1
1	1	0

assign f = e?~a:1'bz;

C-Code Example
```
while(save[i]==k)
  i+=1;
```

```
Loop: ADD r12, r6, r3, LSL #2
      LDR r0, [r12, #0]
      CMP r0, r5
      BNE Exit
      ADD r3, r3, #1
      B Loop
Exit:
```

Logical operations
AND r5, r1, r2 // r5 = r1 & r2
// bitwise AND/OR
ORR r5, r1, r2 // r5 = r1 | r2
MVN r5, r2 // r5 = ~r2
ADD r5, r1, r3, LSL #2 // r5 = r1 + (r3 << 2)
MOV r6, r5, LSR r3 // r6 = r5 >> r3
LDR r5, [r3, r4, LSL #2] // r5 = A[i]
// where r2 is base address, r4 is index

Branch Target Encoding
$PC_{Target} = PC_{Branch} + 8 + 4 * \text{"Target address"}$

Instruction Encoding
COND → usually E
F → instruction format 0 for processing, 1 for transfer
I → if 0, Operand2 is register otherwise immediate
Opcode → specifies operation
S → for branches
Rd → Destination register
Rn → first register operand
Operand2 → second operand

Indirect Branching
Bx rm
↓ equivilant
MOV PC, rm

Post Indexed addressing
STR r3, [r2], #4
↓ equivilant
STR r3, [r2]
ADD r2, r2, #4

Spill/Fill
```
SUB sp, sp, #16
STR r6, [sp, #12]
STR r5, [sp, #8]
STR r4, [sp, #4]
STR r1, [sp, #0]
PUSH {r4-r6, LR}
```
```
LDR r1, [sp, #0]
LDR r4, [sp, #4]
LDR r5, [sp, #8]
LDR r6, [sp, #12]
ADD sp, sp, #16
POP {r4-r6, LR}
```

Arrays
ARRAY:
- Word 0
- Word 1
- ⋮
- Word 9

LDR r1, =ARRAY

Example Encodings
SUB sp, sp, #4 → E24DD004
LDR r8, [sp] → E5905000
STR r10, [sp, #28] → E58DA01C
CMP r2, r3 → E1520003
BLE → DA000001
MVN r0, #0 → E3E00000
MOV PC, LR → E1A0F00E
ADD r6, r2, r5, LRS #1 → E0825CA5
ADD r7, r0, r6, LSL #2 → E0807106
RSB r9, r8, #0 → E2689000

Function Calls
① use r0-r3 for first 4 parameters, Save the rest on stack
② BL to the function
③ Save everything on Stack (stack down)
④ Perform task
⑤ Put result in r0
⑥ mov pc, lr

Conventions
- not expected to save r0-r3, r12
- Expected to save r4-r11

Slide Set 9

Slide Set 10

How? → **Problems** → **Solutions**

① Infinite loop, interrupt always asserted in load IR state → Add a Mask register we know theres an interrupt now execute
② Return address, how to get back to "i+1" → Can't use lr, so use new register LR_IRQ
③ Save, what you need when you return to "i+1" → Can't save on stack, so use a separate stack (location in memory)
④ Stop interrupt, how to tell I/O device we're done → read/write a memory mapped register
⑤ CMP, need to save Status Flags also → Save Status register on interrupt stack
⑥ return, to "i+1" → Set PC to "i+1", restore status flags, unmask, SUBS pc, lr, #4
⑦ Multiple I/O's, more than one interrupt → Different processor modes, Interrupt controller

Interrupt Service Routine
① Recieve interrupt signal at instruction i
② Finish i, then move PC to first instruction of ISR
③ return to "i+1"

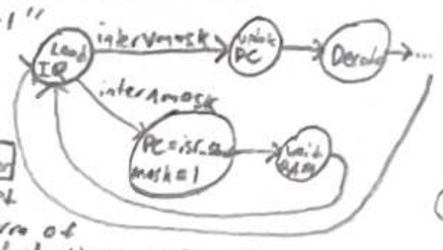
Memory Mapped Registers
- Not in Reg File, these are registers associated with I/O device stored in memory

Polling
- Do nothing until a valid I/O interrupt comes

ANDS r4, r4, #1
// AND r4 and 1 and update status flags. Usually only CMP updates Status flags

Register Banks
- Different modes
- Blue is the same exact physical register
- orange is a different physical register

CPSR → Current Program Status register



Ex Vs interrupt
- interrupt is one form of exception but there are other types ie: memory violation, overflow

Exception Handling
- Exception occurs, Copies CPSR into SPSR

Edge vs Level Sensitive
Level: notify I/O device you are done, ie, read keyboard character
Edge: notify I/O and reset flag in interrupt controller also

Multitasking
- Run A for 20ns, Run B for 20ns, ...
- multithreading, threads under 1 CPU running at same time.
- Multi-Core: threads running on different CPU cores

Slide Set 11

0 1 1 0 0 1 1 1 = 6.4375
2^{-3} 2^{-2} 2^{-1} 2^{0} 2^{1} 2^{2} 2^{3} 2^{4}

Fixed Point

Error
Absolute error = |V(R(x)) − x|
Relative error = |V(R(x)) − x)/x|
where: R(x) is binary 1's and 0's
V(R(x)) is the value of this binary number. x is the actual number

Resolution
- Smallest difference, ie: 2^{-4} for Block (above)
- Absolute error <= ½ resolution

accuracy
- max error over input range you can get

Absolute = max(|V(R(x)) − x|)
Relative = max(|V(R(x)) − x)/x|)

Floating Point
$V = m \cdot 2^{e-x}$
where: m is mantissa
e is exponent
x is bias

Instructions
FLDS → 32 bit FP load
FLDD → 64 bit FP load
FSTS → 32 bit FP load
last S or D means double or single precision
FADDS → 32 bit Addition
FSUBS → 32 bit Subtract
FMULS → 32 bit multiply
FDIVS → 32 bit division

Slide Set 12

x is n times faster than y (speedup)

$n = \frac{\text{performance}(x)}{\text{performance}(y)}$

$n = \frac{\text{execution}(y)}{\text{execution}(x)}$

Iron Law
Execution time = (instruction Count)(CPI)(Cycle time)
= (Cycle count)(cycle time)
$CPI = \frac{\text{Cycle Count}}{\text{Instruction Count}}$

Memory
SRAM: uses 6 MOS transistors to store a bit, slow with large systems
DRAM: uses 1 transistor and a capacitor to store a bit. Cheaper than SRAM, longer than SRAM (need to refresh due to capacitor leaking and keep power on)
Disk Tape: Cheapest
Flash: quantum tunnel in bit

Slide Set 13

Temporal locality:
- Use it once likely to use it again

Spatial locality:
- use it, likely to use addresses around it as well so take those too

Cache on Write
① write through, store both cache and memory on every byte. Conservative and slow
② write back, Store cache only. fast, must mark block as dirty so you know its not in memory

Pipeline Hazards
① Structural, two or more instructions want the same hardware
Sol'n → second instruction has to wait
Sol'n → replicate the hardware block
② Data, can't send info back in time
Sol'n → forwarding, add wires that get info from further back in the pipeline
③ Control, how do you know the next instruction is right
Sol'n → option 1 is Stall, Option 2 is predict branch is not taken If wrong have to correct

IF/ID → ID/EX → EX/MEM → MEM/WB

263.3 in 32-Bit FP

2 | 263	
2 | 131	1
2 | 65	1
2 | 32	1
2 | 16	0
2 | 8	0
2 | 4	0
2 | 2	0
2 | 1	0
2 | 0	1

263: 100000111
0.3: 0100110011.....
SN: $1.00000111 \cdot 2^8$

0.3x2	0.6	0
0.6x2	1.2	1
0.2x2	0.4	0
0.4x2	0.8	0
0.8x2	1.6	1
0.6x2	1.2	1
⋮		0
⋮		0
⋮		1
⋮		1

∴ 263.3
+/− = 0, exp = 10000111, mantissa = 00000111010011...

IEEE 754 32 bit FP → X (1) | xxxxxxxx (8) | xxxx....xxxx (23)
sign | 127+e | mantissa

Cache Organizations
Fully Associative: Comparing all tags is expensive in (time, area, energy)
Direct Mapped: Only one tag to be checked because block only stored in single row. Has more conflict misses than fully associative which can use all rows

Block address: Tag | Index | Block offset
Index = (Block address) % # rows
$\log_2$(block size) removed from block address

Set Associative:
- Index selects set of blocks as opposed to 1. Combo of direct and fully Associative

Write Misses
① write-allocate, if using writeback, read memory into cache
② No-write-allocate, if using write through, Just write straight to memory

Hex → Decimal
$(A\ B\ 0\ 9)_{16} = (43785)_{10}$
$16^3\ 16^2\ 16^1\ 16^0$
$10 \cdot 16^3 + 11 \times 16^2 + 0 + 9 \times 16^0 = 43785$

Decimal → Hex
$(479)_{10} = (1DF)_{16}$
479 ÷ 16 = 29 R15 → F
29 ÷ 16 = 1 R13 → D
1 ÷ 16 = 0 R1 → 1

Adress is 0x10C
Binary → 00010000 100 (index)
tag = 0x10
Block offset = 0x4
Block offset = $\log_2$(block size)

Pipeline Stages
① F → Instruction fetch from memory
② D → instruction decode, read regs in regfile
③ X → execute ALU operation
④ M → access memory LDR or STR
⑤ W → write result into Reg file

Tag/Index/offset
Two Sub caches
index 0
index 1
Search for tag in each subcache. if not found put in subcache

Slide Set 14

$\text{Speedup} = \frac{\text{old Execution time}}{\text{new Execution time}}$

Amdahl's Law

$\text{ExTime}_{new} = \text{ExTime}_{old}[(1 - \text{fractionEnhanced}) + \frac{\text{FractionEnhanced}}{\text{Speedup Enhanced}}]$

Can't be enhanced | Could be enhanced

$\text{Speedup}_{Max} = \frac{1}{1 - \text{fractionEnhanced}}$
Best case possible

Problem - Both threads try to access global variable at the same time.
Sol'n → Synchronization, restrict access until thread 1 is done what it is doing

GPU = Graphics Processing Unit

State Machine Controller

Instructions

~~Case 2: opcode = 110~~
~~op = 10~~

How to Execute each instruction

① Put Sximm8 into Rn

How:

- Set vsel to 0100
- Set write to 1
- set nsel to 100

} S1a ③

② Shift Rm based on the [2:0] Shift and put into RD ④

How:

- Set nsel to 001
- set load a to 0
- set load b to 1

} S2a

- set bsel to 0
- Set asel to 1
- Set load c to 1
- set load s to 1

} S2b

- Set write to 1
- set vsel to 0001

} S2c

③ Add shifted Rm with Rn and put in Rd

How:

- Set nsel to 001
- set load a to 0
- set load b to 1

} S3a

- Set nsel to 100
- set load a to 1
- set load b to 0

} S3b

- set asel to 0
- set bsel to 0
- Set load c to 1
- set load s to 1

} S3c

- Set write to 1
- Set vsel to 0001
- Set nsel to 010

} S3d

Top level:

`define S1 110

① `define S1a 10
//Execute R[Rn] = sx(im8)

② `define S1b 00
// Execute R[RD] = sh_Rm

`define S2 101

`define S2a 00
//Execute R[Rd] = R[Rn] + sh_Rm

④ `define S2b 01
//Execute Status = f(R[Rn] - sh_Rm)

⑤ `define S2c 10
//Execute R[Rd] = R[Rn] & sh_Rm

⑥ `define S2d 11
//Execute R[Rd] = ~sh_Rm

⑤ AND shifted Rm and Rn and put in Rd

How:

- Set nsel to 001
- Set load a to 0
- set load b to 1

} S5a

- Set nsel to 100
- set load a to 1
- set load b to 0

} S5b

- Set asel to 0
- set bsel to 0
- set load c to 1
- set load s to 1

} S5c

- set nsel to 010
- set vsel to 0001
- set write to 1

} S5d

⑥ NOT shifted Rm and put in Rd

How:

- set nsel to 001
- set load a to 0
- set load b to 1

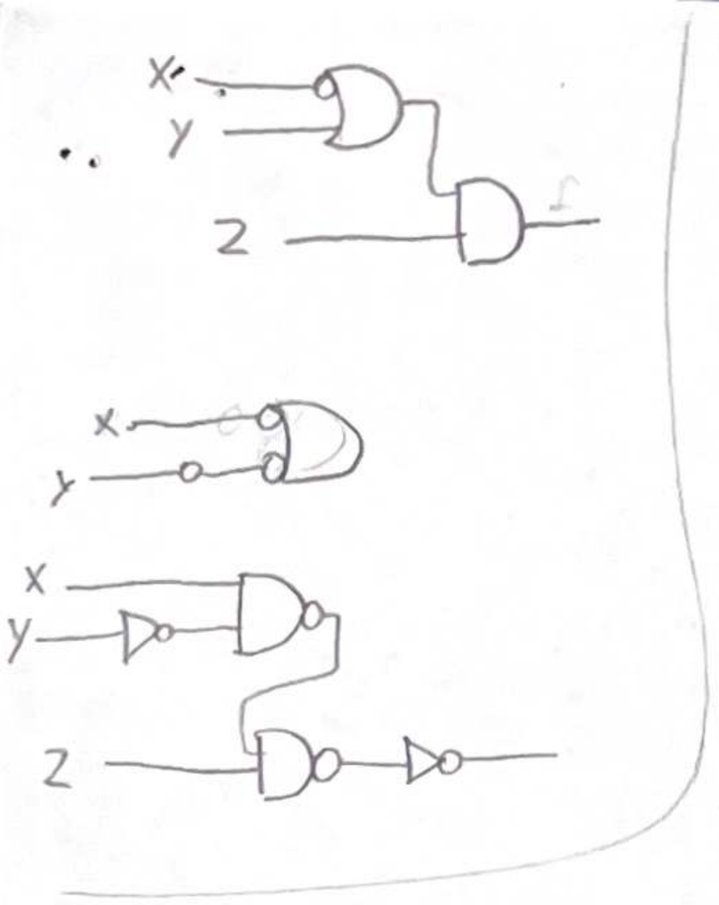
} S6a

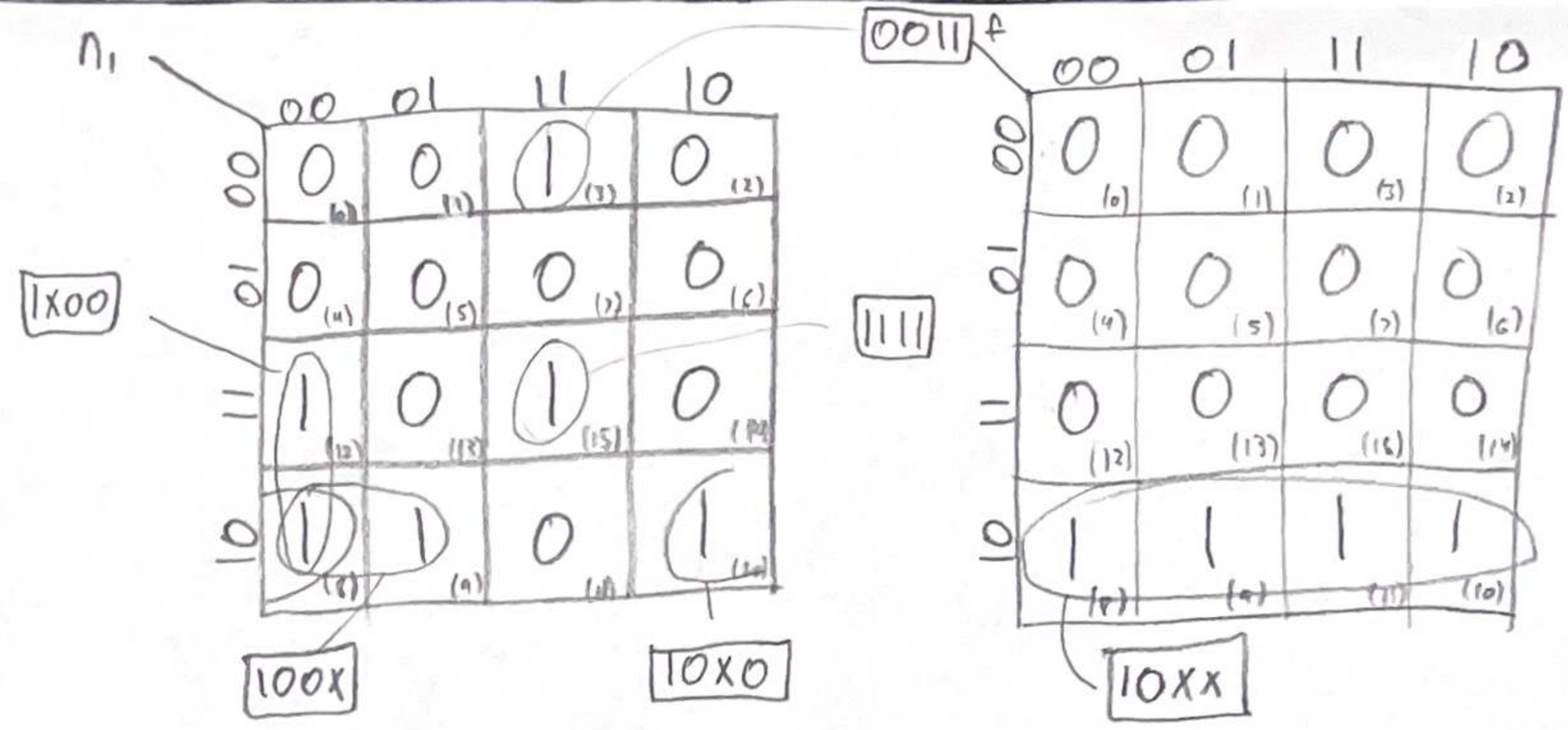
- Set asel to 1
- set bsel to 0
- Set load c to 1
- set load s to 1

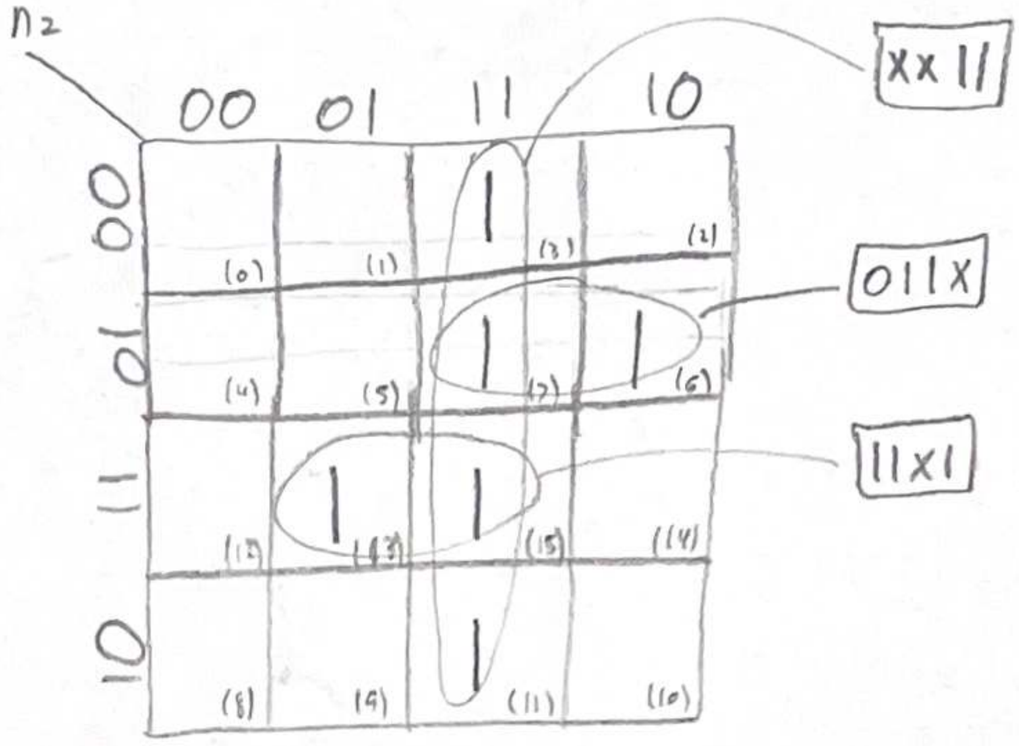
} S6b

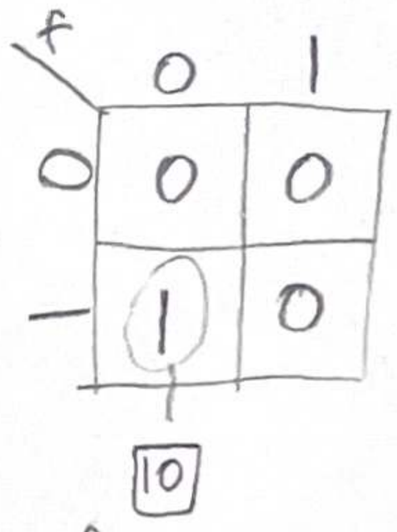
- set nsel to 010
- set vsel to 0001
- set write to 1

} S6c









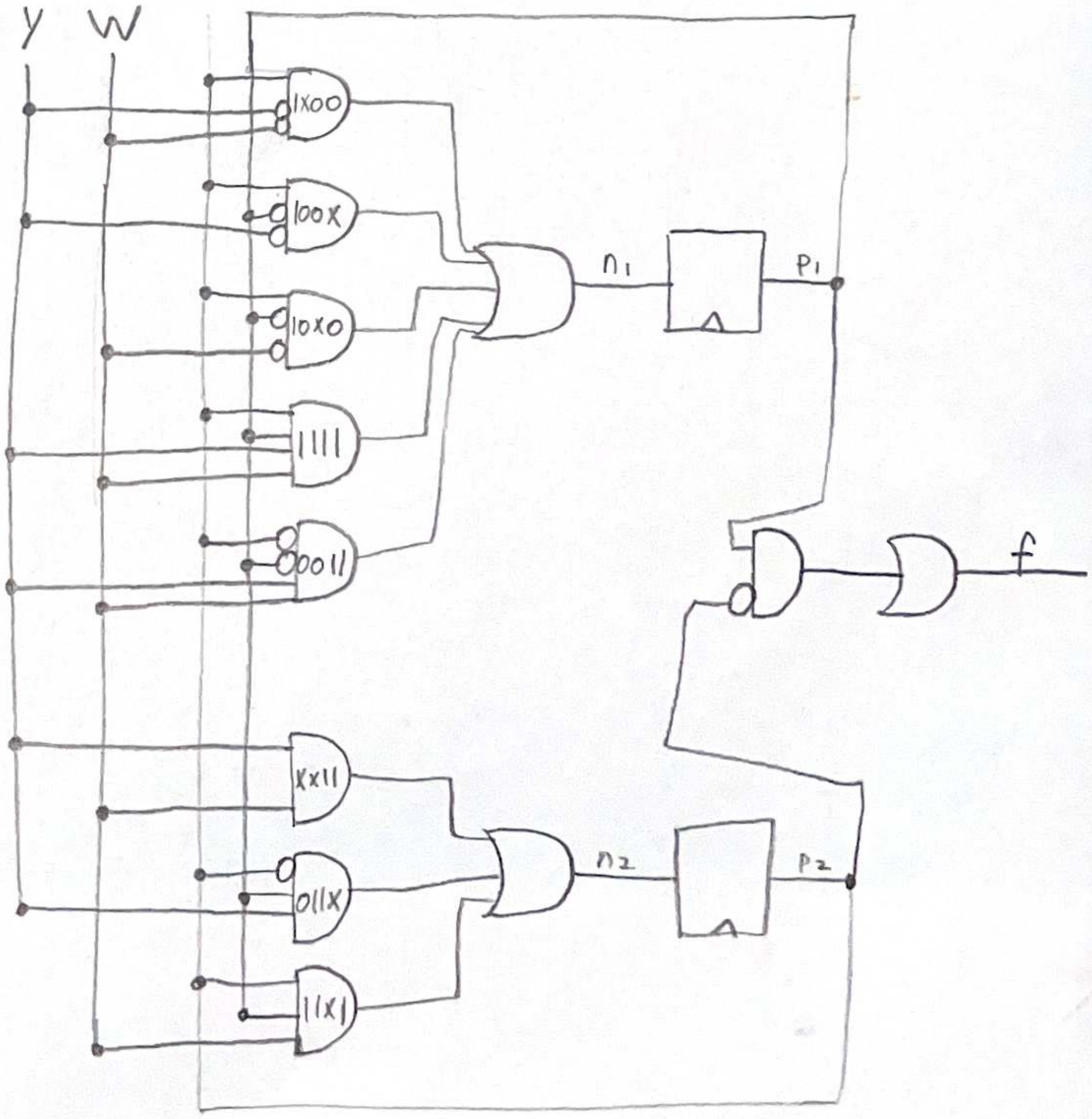
$f = p_2 \wedge \neg p_1$

State encoding

A = 00
B = 01
C = 11
D = 10

p_2, p_1 = present state
n_2, n_1 = next-state
y, w = in
f = out

	p_2	p_1	y	w	n_2	n_1	f	
A	0	0	0	0	0	0	0	(0)
	0	0	0	1	0	0	0	(1)
	0	0	1	0	0	0	0	(2)
	0	0	1	1	1	1	0	(3)
B	0	1	0	0	0	0	0	(4)
	0	1	0	1	0	0	0	(5)
	0	1	1	0	1	0	0	(6)
	0	1	1	1	1	0	0	(7)
D	1	0	0	0	0	1	1	(8)
	1	0	0	1	0	1	1	(9)
	1	0	1	0	0	1	1	(10)
	1	0	1	1	1	0	1	(11)
C	1	1	0	0	0	1	0	(12)
	1	1	0	1	1	0	0	(13)
	1	1	1	0	0	0	0	(14)
	1	1	1	1	1	1	0	(15)



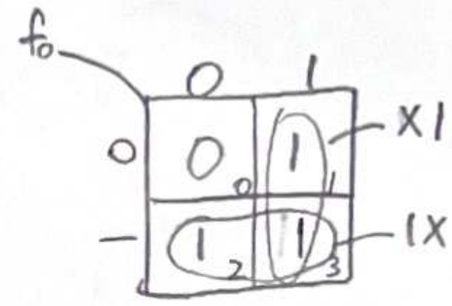
State encoding

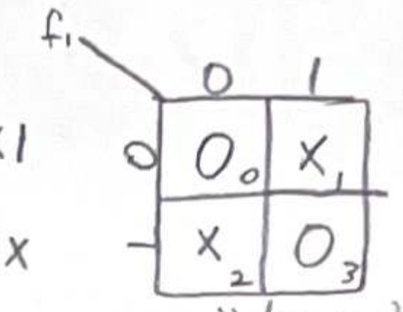
A = 00
B = 01
C = 11
D = 10

let:

p_1, p_0 = present state
n_1, n_0 = next state
X_2, X_1 = inputs
f_1, f_0 = outputs

p_1	p_0	f_1	f_0	
0	0	0	0	(0)
0	1	1	X	(1)
1	0	1	X	(2)
1	1	1	0	(3)

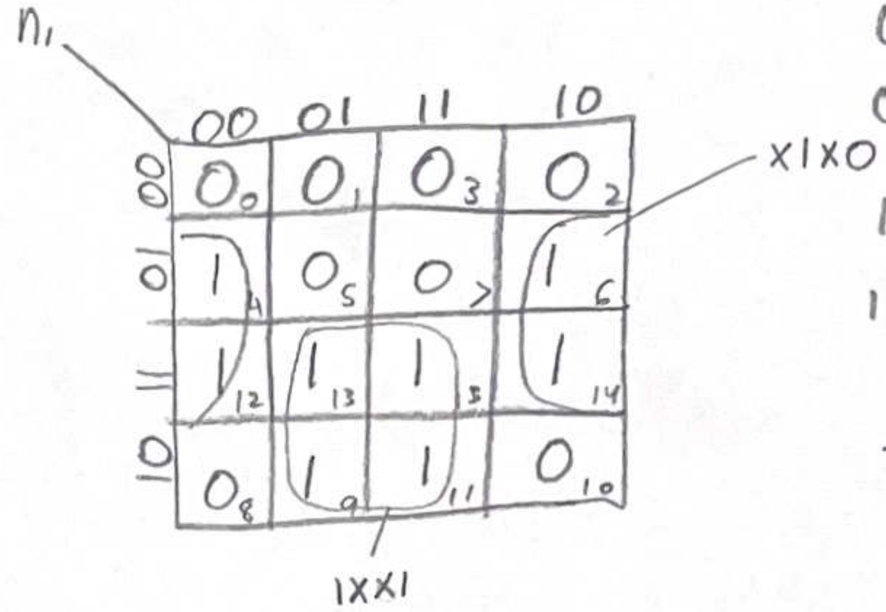




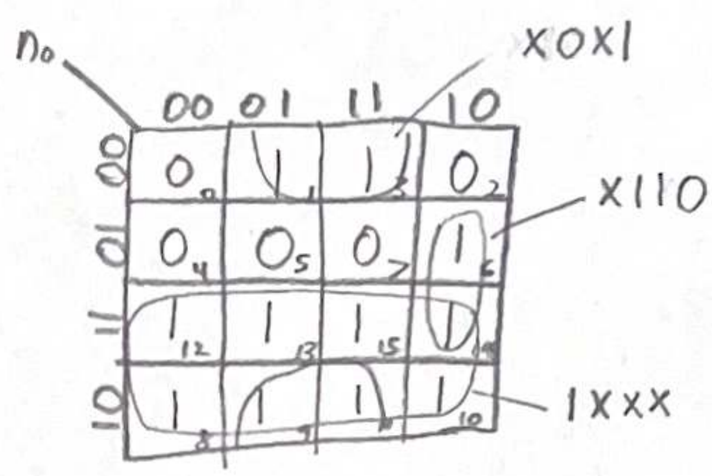
"doesn't depent on present

p_1	p_0	f_0
0	0	0
0	1	1
1	0	1
1	1	1

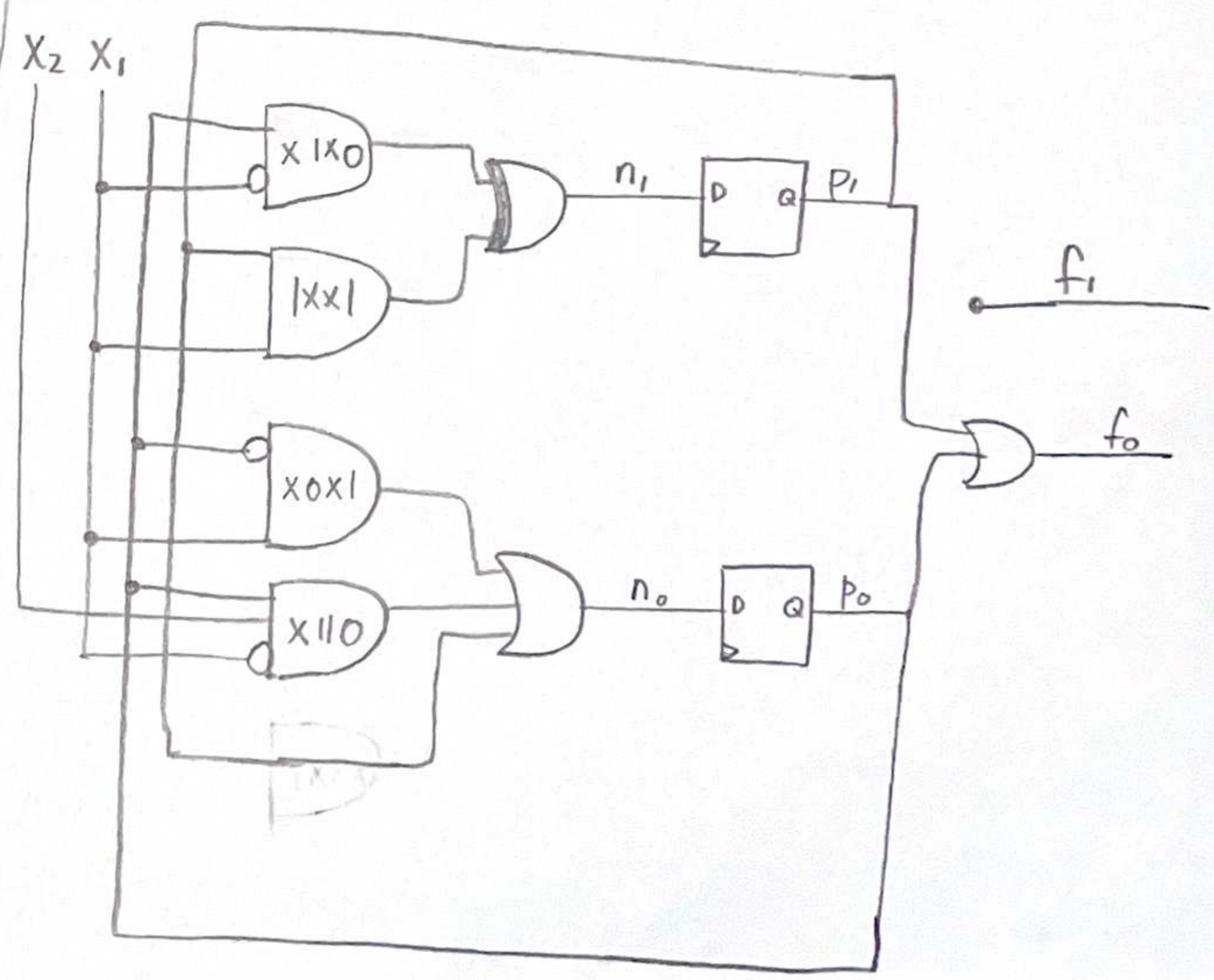
this is an OR Gate



	p_1	p_0	X_2	X_1	n_1	n_0	
A	0	0	0	0	0	0	(0)
	0	0	0	1	0	1	(1)
	0	0	1	0	0	0	(2)
	0	0	1	1	0	1	(3)
B	0	1	0	0	1	0	(4)
	0	1	0	1	0	0	(5)
	0	1	1	0	1	1	(6)
	0	1	1	1	0	0	(7)
D	1	0	0	0	0	1	(8)
	1	0	0	1	1	1	(9)
	1	0	1	0	0	1	(10)
	1	0	1	1	1	1	(11)
C	1	1	0	0	1	1	(12)
	1	1	0	1	1	1	(13)
	1	1	1	0	1	1	(14)
	1	1	1	1	1	1	(15)



0001
0011
1001
1011

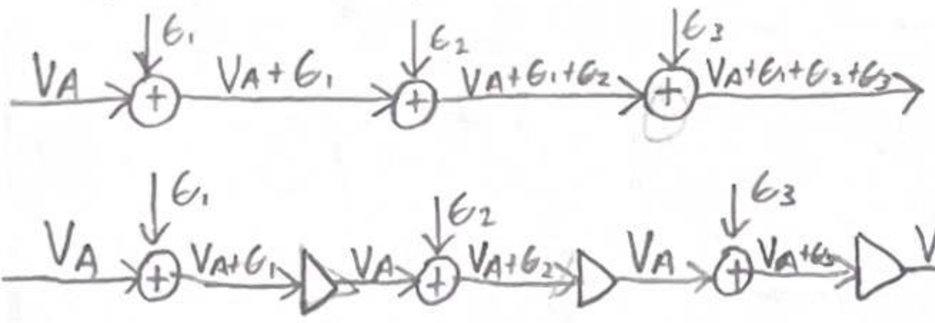


CPEN 211

Chapter 1: Digital Abstraction

Noise

ANALOG (Noise accumulates)



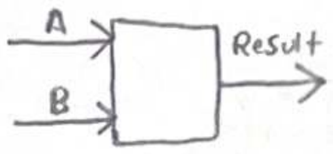
DIGITAL

▷ Buffer: determines if its in a 0 or 1 range and makes it a pristine 0 or 1. Effectively scrubbing noise.

Digital Signals

- N elements requires $\log_2 N$ bits, ie: 8 colours needs $\log_2 8 = 3$ bits

Combinational Logic Circuit



- no memory
- result determined by present inputs

output = f(input)

Verilog

- Verilog is a hardware description language HDL NOT code/software
- Quartus does not run verilog it runs the logic gates made from the verilog

Sequential Logic Circuit

- Has memory

Chapter 3: Boolean Algebra

Axioms

$1 \wedge x = x \qquad 0 \wedge x = 0$

$1 \vee x = 1 \qquad 0 \vee x = x$

$\bar{0} = 1 \qquad \bar{1} = 0$

Equivilantly

$\neg 0 = 1 \qquad \neg 1 = 0$

Properties

Commutative: $x \wedge y = y \wedge x$

Associative: $x \wedge (y \wedge z) = (x \wedge y) \wedge z$

Distributive: $x \wedge (y \vee z) = (x \wedge y) \vee (y \wedge z)$

Idempotence: $x \wedge x = x$; $x \vee x = x$

Complementation: $x \wedge \bar{x} = 0$; $x \vee \bar{x} = 1$

Absorbtion: $x \wedge (x \vee y) = x$

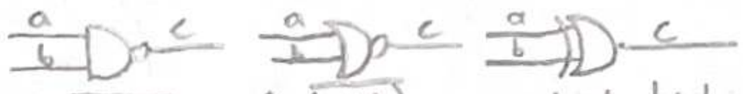
Combining: $(x \wedge y) \vee (x \wedge \bar{y}) = x$

De Morgans: $\overline{(x \wedge y)} = \bar{x} \vee \bar{y}$

Logic Gates

AND OR NOT

NAND NOR XOR

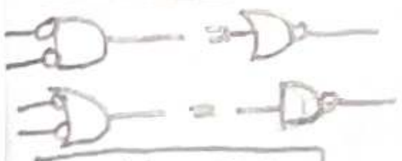


$f = \overline{(a \wedge b)}$ $\quad f = \overline{(a \vee b)}$

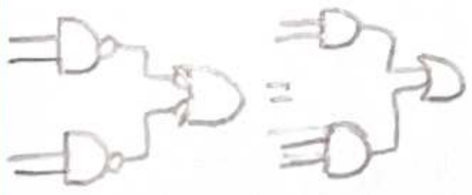
- output high if one input is high
- output true if odd number of inputs is high

a	b	XOR
0	0	0
1	0	1
0	1	1
1	1	0

De Morgan



Bubble Rule



Truth Tables

a	b	$a \wedge b$	$a \vee b$
0	0	0	0
1	0	0	1
0	1	0	1
1	1	1	1

a	$\bar{a}$
0	1
1	0

Dual Functions

- Swith everything

$f(x,y) = (x \wedge 1) \vee (0 \vee \bar{y})$

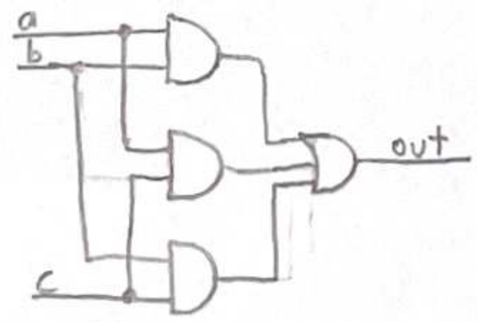
$f^D(x,y) = (x \vee 0) \wedge (1 \wedge \bar{y})$

Verilog

AND OR NOT XOR

& | ~ ^

```
module Majority(a,b,c,out);
  input a,b,c;
  output out;
  assign out = (a&b)|(a&c)|(b&c);
endmodule
```



Duality

- You can change every $\wedge \to \vee$ and $\vee \to \wedge$ and $0 \to 1$ and $1 \to 0$ and everything holds

Normal Form

a	b	c	q
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$f(a,b,c) = (\bar{a} \wedge b \wedge c) \vee (a \wedge \bar{b} \wedge c) \vee (a \wedge b \wedge \bar{c}) \vee (a \wedge b \wedge c)$

Chapter 6: Combinational Logic

Ɛ

-no memory

Binary

2^n = combos
where n is bits

$2^3 = 8$
$2^4 = 16$

Karnaugh Map

	00	01	11	10
00	0	1	1	1
01	0	1	1	0
11	0	1	0	0
10	0	0	1	0

closed

-closed under acyclic composition
-you can add them up and its still combinational as long as you don't make loops

Grouping

-Can group in
8
4
2

- must choose biggest and work down

Prime implicant

-an implicant that cannot be made larger
-unique
-shares no bits

Essential Prime implicant

-an implicant with a 1 only covered once

minterm

-Product term

$f = \sum_{dcba} m(1,2,3,5,7,11,13)$

-must have min 4 if dcba

Normal Form

-Sum of the product terms

Don't Cares

-allows you to cover a larger area with X's and thus make lesse implicants

Implicant

-a minterm could have value 0
-an implicant could have less terms by combining

Chapter 7 Verilog

"To describe combinational logic to be synthesized we only use assign and case statements"

Structure

```
module Count(in, out);
  input [3:0] in;
  output [out;
  <Module Body (logic)>
endmodule
```

Wire

-outputs are a wire by default
- Assigned by an ASSIGN statement
-Used to connect modules

Reg

-If assigned in a case statement or an always block
-Not a register

Case Statement

-specifys a truth table
-must have all possibilities covered or include a default

Always Block

always@(in) begin
end

always@(*) begin
end

begin/end

-like { } in C
(Brackets)

Assign

Computed whenever RHS changes.
Describes hardware

Testbench

Initial → like an always block
#100 → wait 100ps
$display("like printf");
repeat → loop generator

===
!==

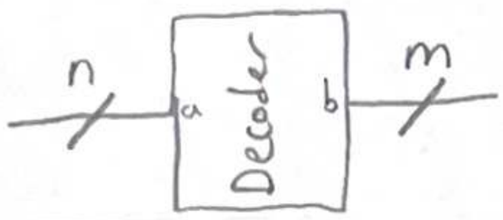
Chapter 8 Combinational Building Blocks

Decoder

- Converts one type of symbol to a different type

In general:

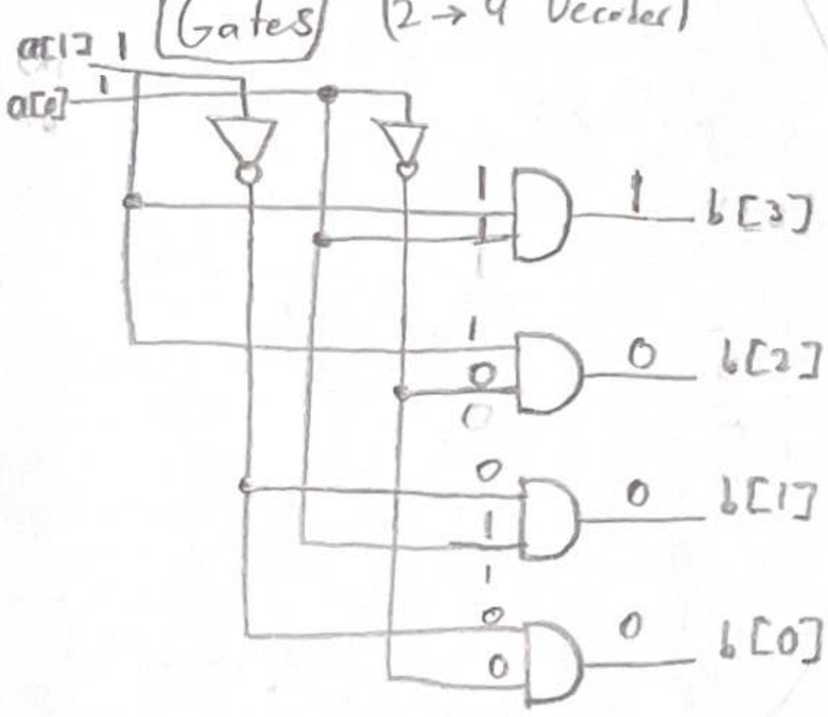
Decoder: Binary → One-hot
Encoder: one-hot → binary



Verilog (Decoder)

```
// n -> m decoder
// a - binary input (n bits wide)
// b - one-hot output (m bits wide)
module Dec(a,b);
    parameter n=2;
    parameter m=4;

    input [n-1:0] a;
    output [m-1:0] b;

    assign b= 1<<a;
endmodule
```

Gates (2 → 4 Decoder)



Key Building Blocks

1. Decoder
2. Encoder
3. Multiplexer
4. Arbiter
5. Comparator
6. Read only memories

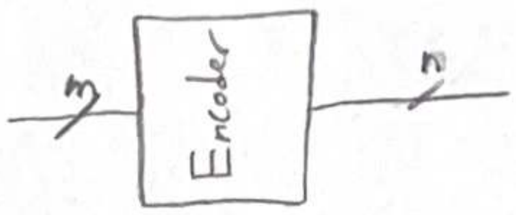
Paramater

- Can override default values

```
Dec #(3,8) dec38 (a,b);
```

Encoder

- Converts one-hot to binary

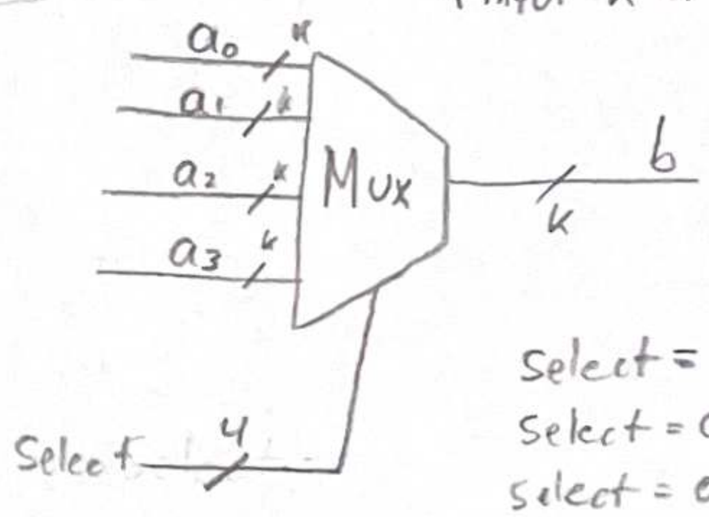



```
module Enc42(a,b);
    input [3:0] a;
    output [1:0] b;

    assign b= {a[3]|a[2], a[3]|a[1]};
endmodule
```

Multiplexer

- Chooses what input becomes the output

4 input k bit mux, one-hot select

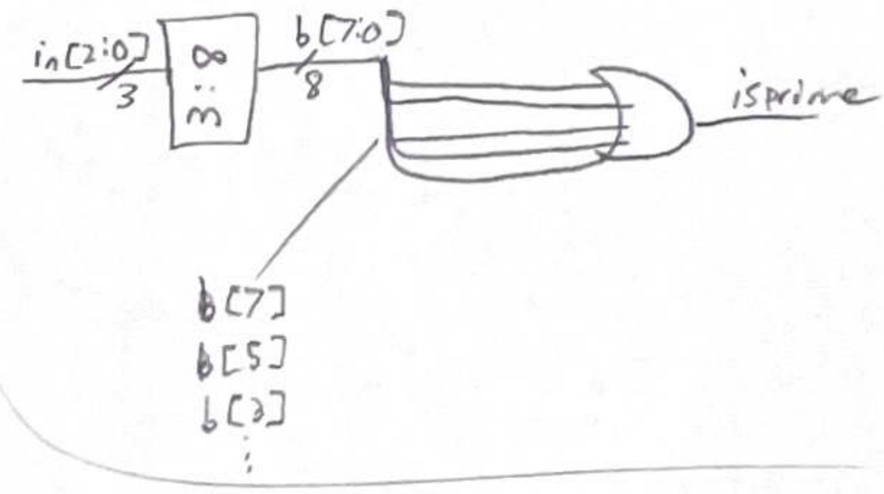


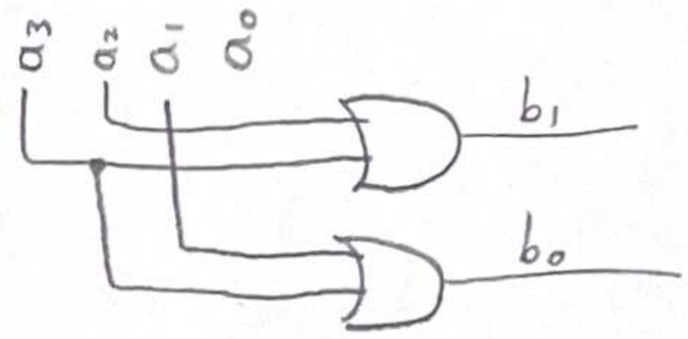
Select = 0001	b = a0
Select = 0010	b = a1
Select = 0100	b = a2
Select = 1000	b = a3

```
module Mux4(a3,a2,a1,a0,s,b);
    parameter k=1;
    input [k-1:0] a0,a1,a2,a3;
    input [3:0] s;
    output [k-1:0] b;
    wire [k-1:0] b;

    assign b = ({k{s[0]}} & a0 |
               ({k{s[1]}} & a1 |
               ({k{s[2]}} & a2 |
               ({k{s[3]}} & a3);
endmodule
```

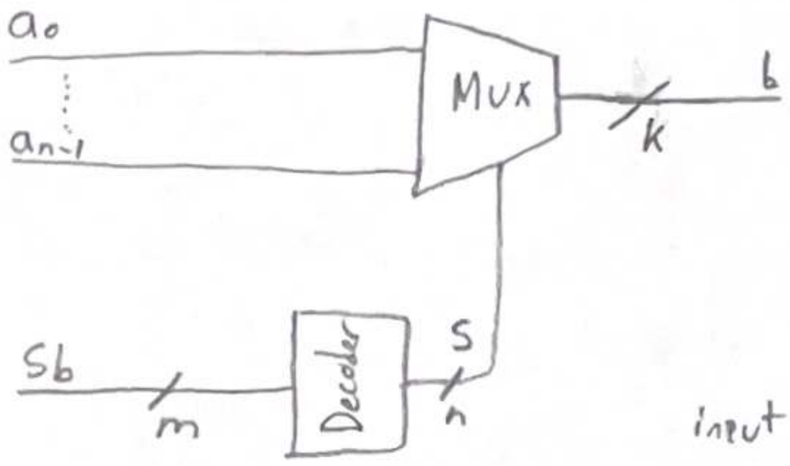
Example (Prime number function)

```
module Primed(in, isprime);
    input [2:0] in;
    output      isprime;
    wire [7:0] b;

    wire isprime = b[1] | b[2] | b[3] | b[4] | b[5] | b[6] | b[7];

    Dec #(3,8) d(in,b);
endmodule
```





Binary-Select Mux



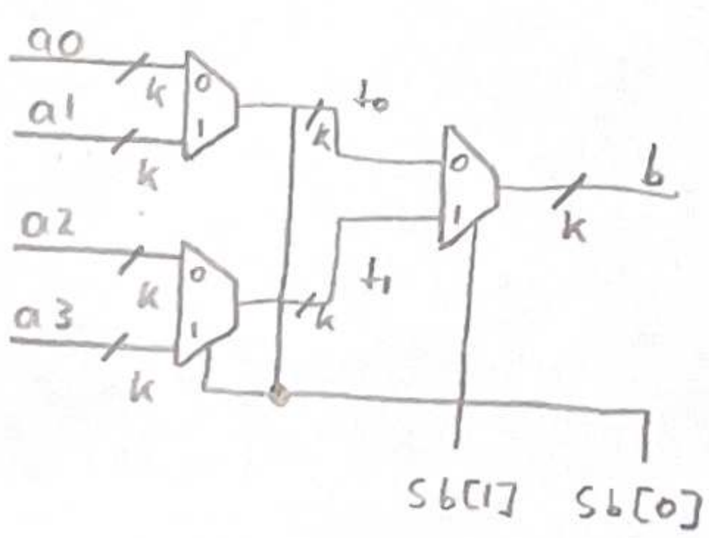
-Less efficient

```
module Muxb3(a2,a1,a0,Sb,b);
  parameter k=1;
  input [k-1:0] a2,a1,a0
  input [1:0] Sb;
  output [k-1:0] b;
  wire [2:0] s;

  Dec #(2,3) d(Sb,s);
  Mux3 #(k) m(a2,a1,a0,s,b);
endmodule
```

Factoring

4:1 Mux Binary Select
Built out of 2 smaller Muxes

```
module Mux4b(a3,a2,a1,a0,sb,b);
  parameter k=1;
  input [k-1:0] a3,a2,a1,a0;
  input [1:0] sb;
  output [k-1:0] b;
  wire [k-1:0] t0, t1;

assign t0 = sb[0] ? a1 : a0;
assign t1 = sb[0] ? a3 : a2;
assign b = sb[1] ? t0 : t1;
```



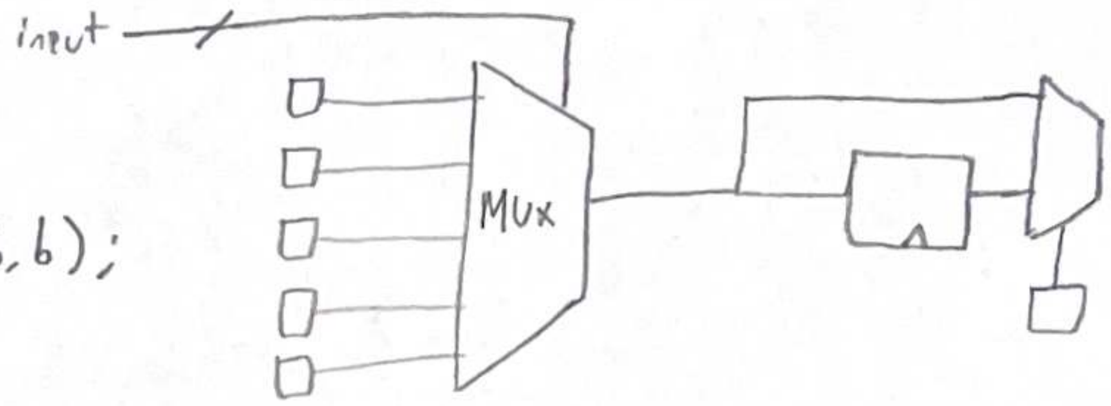
Shannon Expansion

$F(X_0, X_1, X_2) = (\overline{X_0} \wedge F(0, X_1, X_2)) \vee (X_0 \wedge F(1, X_1, X_2))$

-used to simplify muxes to smaller muxes

-Can be done with any X input

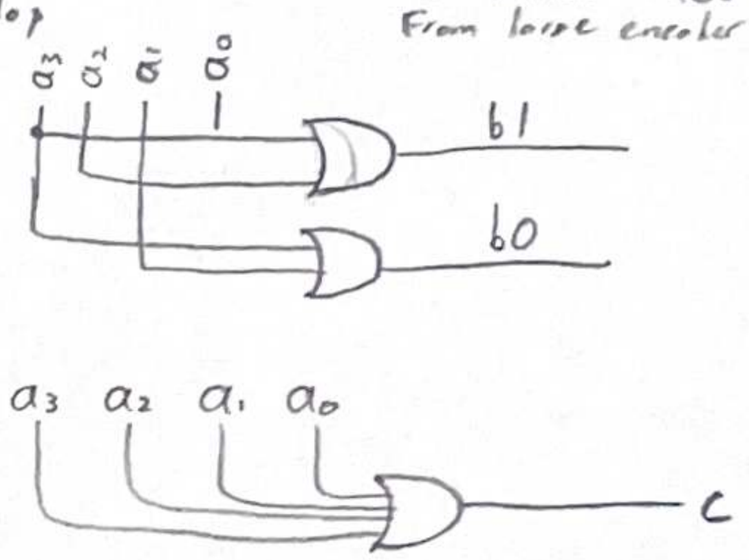
FPGA Logic Block



lookup table

Flip Flop

*C values = Blue
*D values = Red
From large encoder example



4:2 Encoder

```
module Enc42(a,b,c);
  input [3:0] a;
  output [1:0] b;
  output c;
  wire [1:0] b;
  wire c;

  assign b[1] = a[3] | a[2];
  assign b[0] = a[3] | a[1];
  assign c = |a;   // c = a[0]|a[1]|a[2]|a[3];
endmodule
```

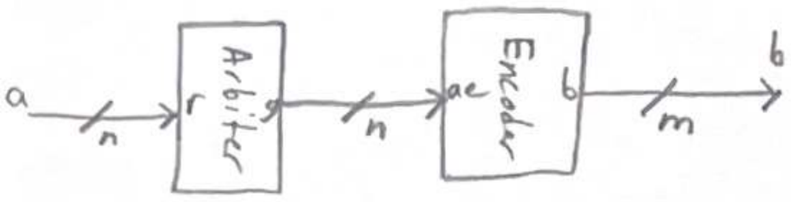
Arbiter

-Finds first 1 in r starting at LSB and outputs a 1-hot-code as the position of that one

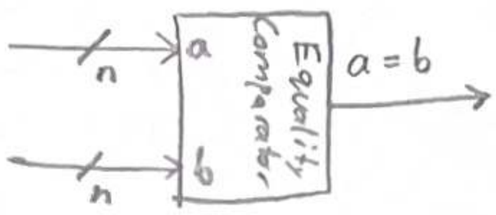
Priority Encoder

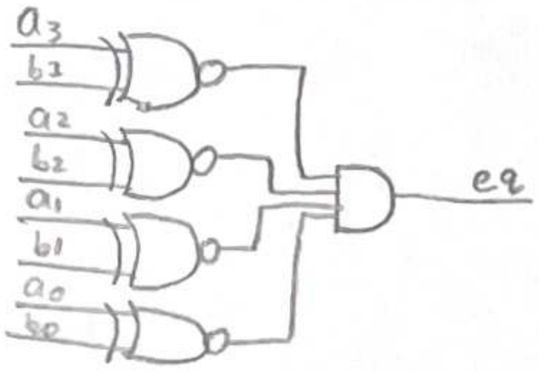
- Finds the first occurrence of a 1 in the input bus starting at the LSB and outputs its position in binary

```
module PriorityEncoder83(r,b);
    input [7:0] r;
    output [2:0] b;
    wire [7:0] g;
    Arb #(8) a(r,g);
    Enc83    e(g,b);
endmodule
```



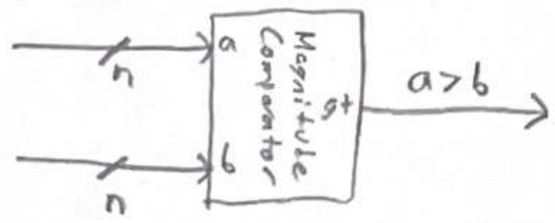
Equality Comparator





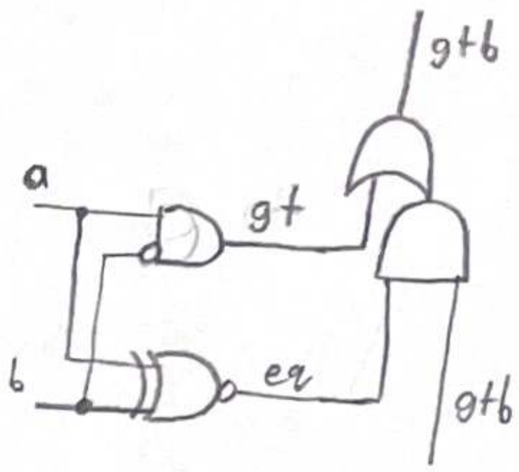

```
module EqComp(a,b,eq);
    parameter k=8;
    input [k-1:0] a,b;
    output eq;
    wire eq;

    assign eq = (a==b);
endmodule
```

Magnitude Comparator




```
module MagComp(a,b,gt);
    parameter k=8;
    input [k-1:0] a,b;
    output gt;
    wire[k-1:0] eqi = a~^b;
    wire [k-1:0] gti = a & ~b;
    wire [k:0]  gtb = {((eqi[k-1:0] & gtb[k-1:0]) | gti[k-1:0]), 1'b0};
    wire gt = gtb[k];
```

a>b iff a[i]>b[i] AND a[j] = b[j] for j>i

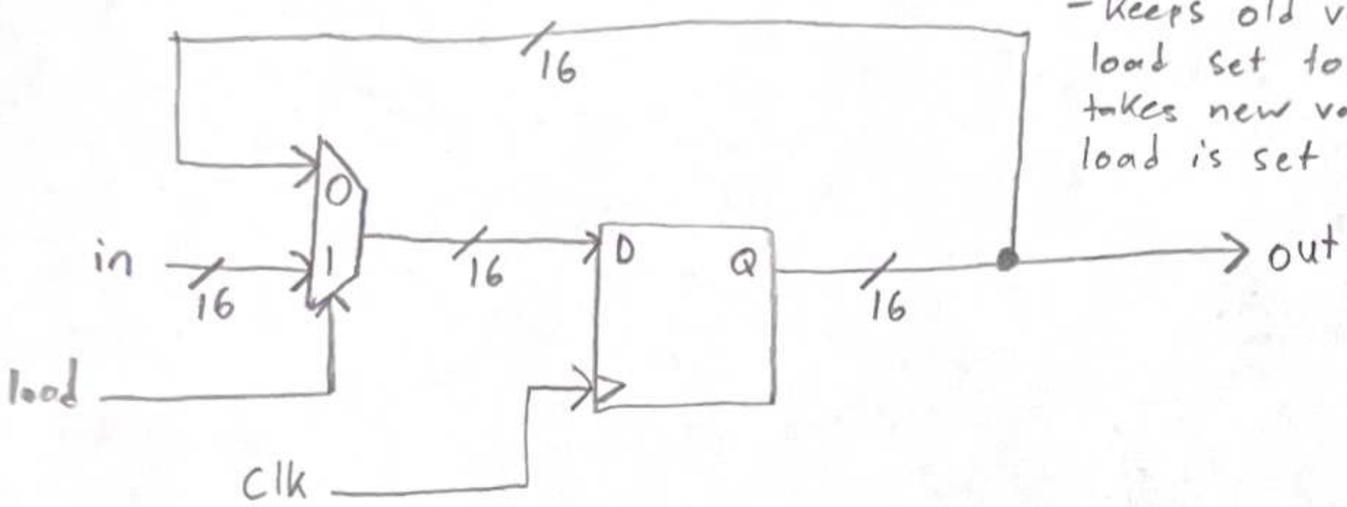



```
module MagComp(a,b,gt);
    parameter k=8;
    input [k-1:0] a,b;
    output gt;
    wire gt = (a>b);
endmodule
```

Read only Memory (ROM)

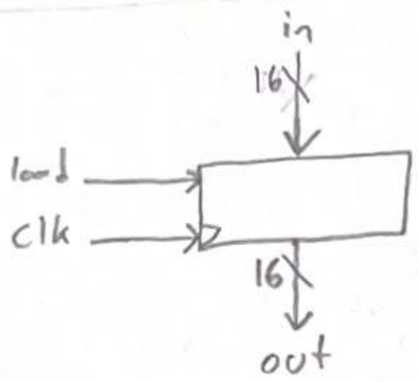
```
module ROM(read_address, dout);
    parameter data_width = 32;
    parameter addr_width = 4;
    parameter filename = "data.txt";
    input [addr_width-1:0] read_address;
    output [data_width-1:0] dout;
    reg [data_width-1:0] mem[2**addr_width-1:0];

    initial $readmemb(filename, mem);
    assign dout = mem[read_address];
endmodule
```

Register with load enable



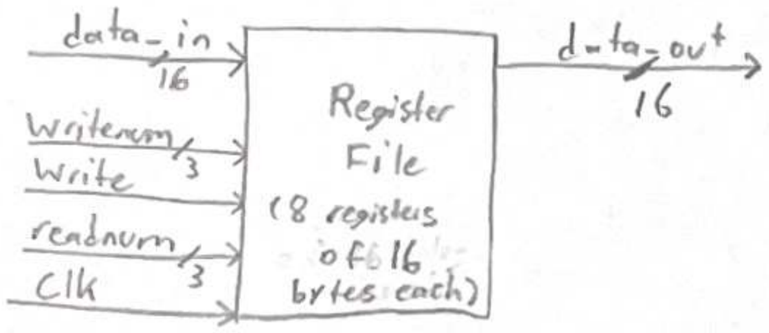
- 16 Flip Flops
- Remembers
- Keeps old value if load set to 0 and takes new value if load is set to 1

- Remebers value as long as load set to 0

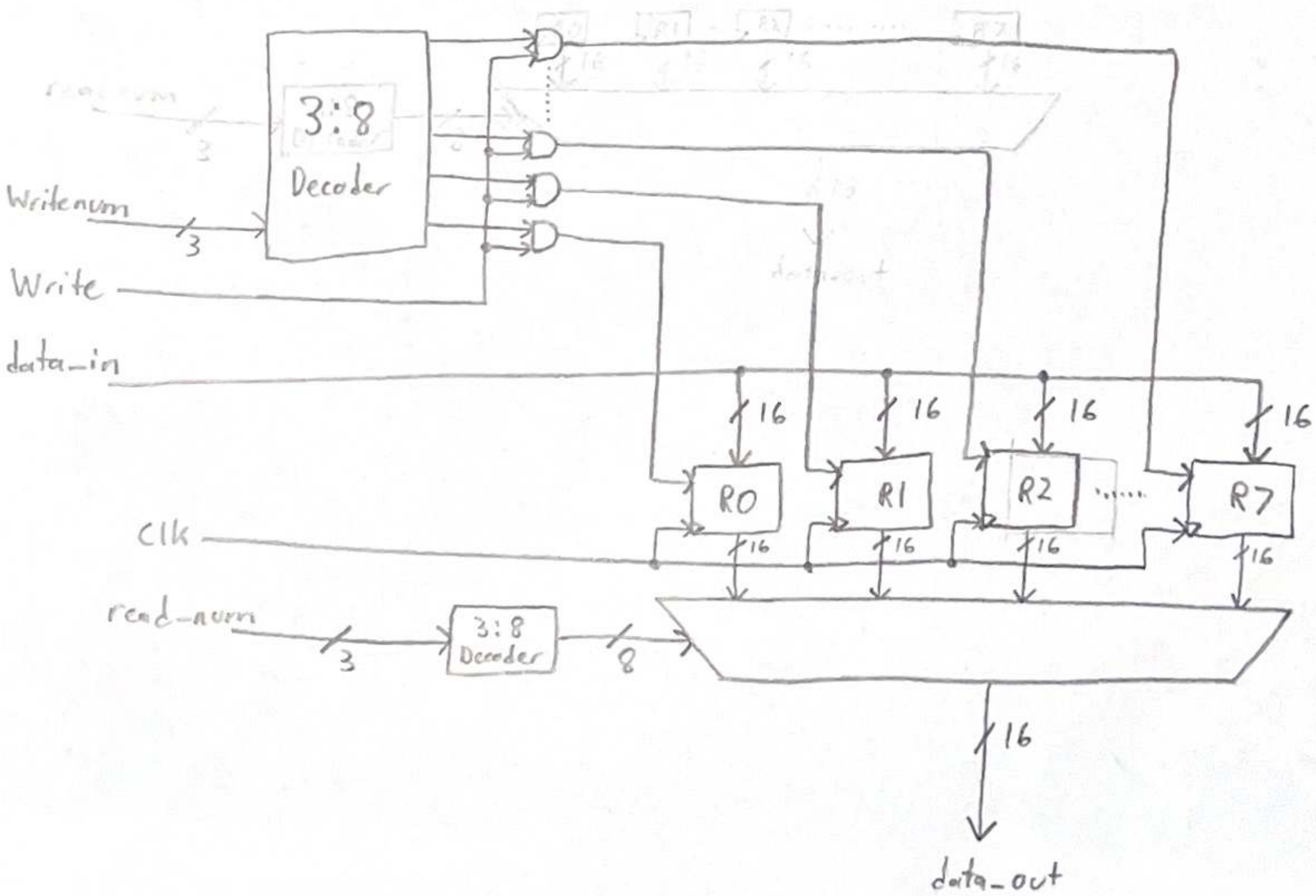



```
module vDFFE(clk, en, in, out);
    parameter: n = 1; // Width
    input clk, en;
    input [n-1:0] n;
    output [n-1:0] out
    reg   [n-1:0] out
    wire  [n-1:0] next_out;

    assign next_out = en? in:out;

    always @(posedge clk)
        out= next_out;
endmodule
```

Register File



- data-in, data-out is values
- Writenum, readnum are addresses



ARM

Levels of Representation

High-Level
```
temp=V[k]
temp++;
```

Assembly
```
LDR R5, [R2]
LDR R6, [R2,#4]
```

Machine language
```
1100 0101
1111 1000
```

Control signals ALUop[0:3] ← f(instr Reg[25:20])

Assembly Syntax

Ex C: a=b+c+d+e;

ARM:
```
ADD a,b,c //add b and c and place in a
ADD a,a,d
ADD a,a,e
```

Ex C: f=(g+h)-(i+j);

ARM:
```
ADD t0, g, h
ADD t1, i, j
SUB f, t0, t1 // f = t0-t1
```

Types of Memory

1. DRAM (Dynamic)
2. SRAM (static)
3. Flash memory (holds value even when powered off)

Reading from Memory

```
LDR R5, [R2, #8]
```
address

// R2 contains <val> 100, takes that and stores it in R5

#(<val>) is the offset to get to the right address

Bytes

- int takes up 32 bit 'word' broken up into 4-8bit bytes

0000 0000 0000 1010

- ARM puts LSB first in 32 bit 'word' "little endian"
- Could also put MSB first "big endian"

Address	Data
16	
12	
8	
4	10
0	

Writing to Memory

```
STR r5, [r3, #48]
```
// stores the value in r5 at that address

Immediate operands

```
ADD r3, r3, #4
//r3=r3+4
```

Instruction Field

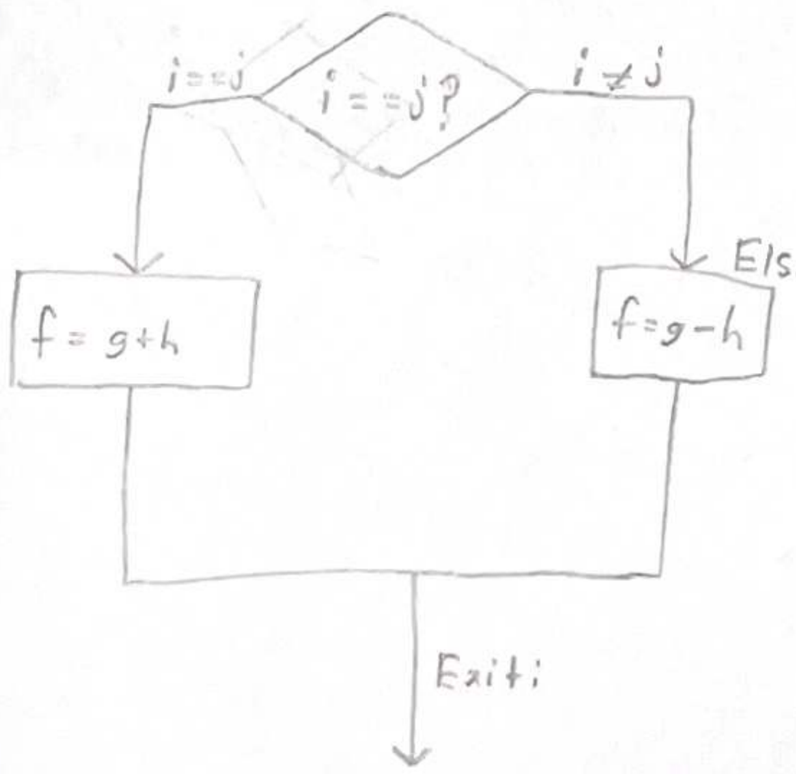
	Cond	F	I	Opcode	S	Rn	Rd	Operand 2
# bits	4	2	1	4	1	4	4	12

Logical Operations

```
AND r5, r1, r2 // reg r5 = reg r1 & reg r2
ORR r5, r1, r2 // reg r5 = reg r1 | reg r2
MVN r5, r2 // reg r5 = ~reg r2
MOV r6, r5 // reg r6 = reg r5
```

Branches

```
if(i==j)
  f=g+h;
else f=g-h;
```
C Code




```
CMP R3, R4
BNE Else //Branch not equal, go to Else:
ADD r0, r1, r2 //skipped if i≠j
B Exit
Else: SUB r0, r1, r2
Exit:
```

Loops

C Code
```
while(save[i]==k)
  i+=1;
```

ARM Assembly
```
Loop: ADD r12, r6, r3, LSL #2
      //r12 = address of save[i]
      LDR r0, [r12, #0]
      //Temp reg r0=save[i]
      CMP r0, r5
      BNE Exit
      ADD r3, r3, #1
      B Loop
Exit:
```

Conditional Execution

- Can eliminate branches

```
CMP R3, R4
BNE Else
ADD r0, r1, r2
 B  Exit
Else: SUB r0, r1, r2
Exit:
```

→

```
CMP R3, R4
ADDEQ r0, r1, r2
SUBNE r0, r1, r2
```

Hexadecimal

D_{10}	H_{16}
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	a
11	b
12	c
13	d
14	e
15	f

Hexidecimal → Decimal:

$(A\ B\ 0\ 9)_{16} = (43785)_{10}$

$\downarrow\downarrow\downarrow\downarrow$

$16^3\ 16^2\ 16^1\ 16^0$

$10 \cdot 16^3 + 11 \cdot 16^2 + 0 \cdot 16 + 9 \cdot 1 = 43,785$

Decimal → Hexadecimal:

$(479)_{10} = (1DF)_{16}$

$479 \div 16 = 29$ $\boxed{R15}$ → F

$29 \div 16 = 1$ $\boxed{R13}$ → D

$1 \div 16 = 0$ $\boxed{R1}$ → 1

Step 1

- ARM uses R0–R3 for the first 4 parameters passed
- More than 4 parameters requires using memory

```
main:
    MOV R0, #1
    MOV R1, #5
    MOV R2, #9
    MOV R3, #20
```

Step 2

- ARM uses branch and link BL to transfer control to subroutine
- Places address to return home to in LR (R14)

```
BL leaf_function
```

Step 5

- ARM puts the result to be returned in R0.

*Note this overwrites Parameter 1 placed in R0. This is ok because we will save a copy of it on the stack

```
MOV R0, R4
```

Indirect Branches

```
Bx  Rm
MOV PC, Rm
```

C Code to Execute

```
/* main.c */
extern int leaf_example(int, int, int, int);
void main(){
    int result = leaf_example(1, 5, 9, 20);
    if (result > 10)
        ⋮
}

/* leaf.c */
int leaf_example(int g, int h, int i, int j){
    int f;
    f = (g+h) - (i+j);
    return f;
}
```

Step 6

```
MOV pc, lr
```

- PC (R15) in ARM
- Value in lr is pc adress at BL+4

Function Call in ARM

Steps:

1. Put parameters where function can find them
2. Transfer control to the function
3. *Acquire Storage for function
4. Perform task
5. Put result where calling function can access it
6. Return control to the point of origin (function might be called from many places)

Terminology

Caller: makes the call ie: main

Callee: function being called ie: leaf_example

Convention:

- ARM uses Callee Saving